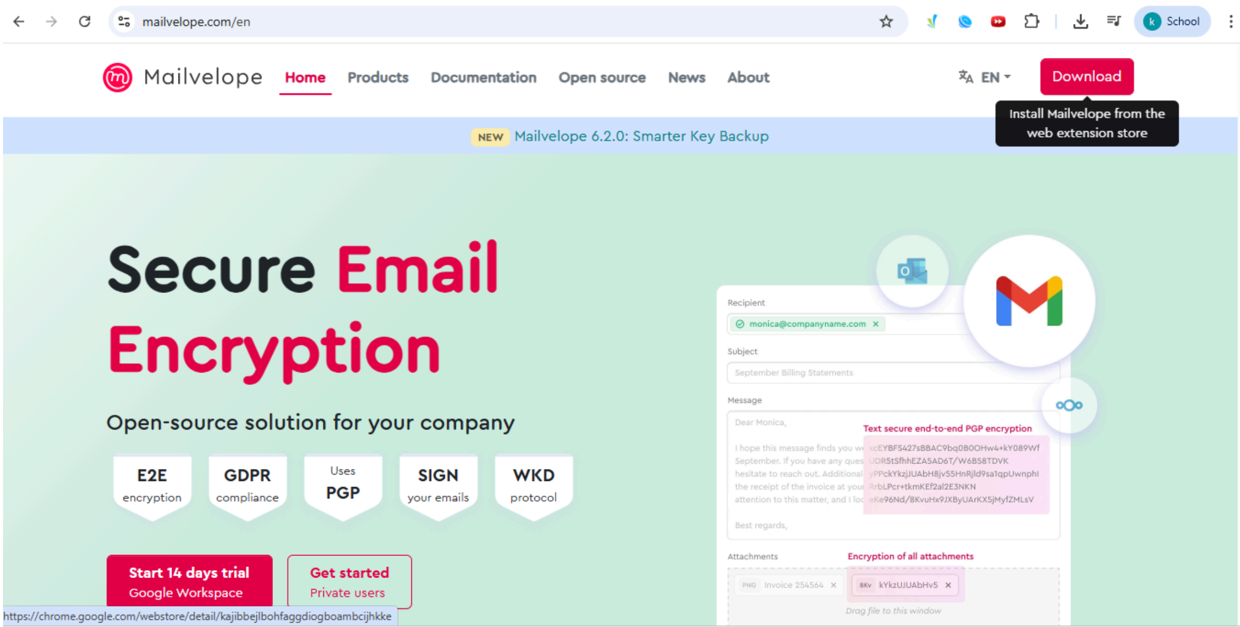
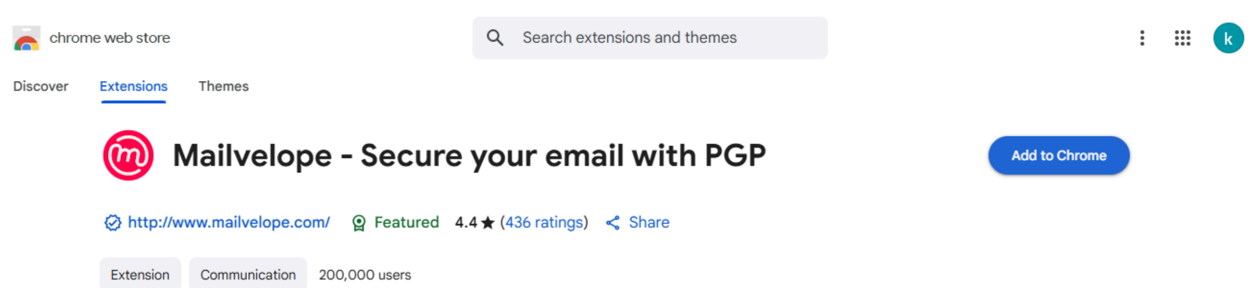
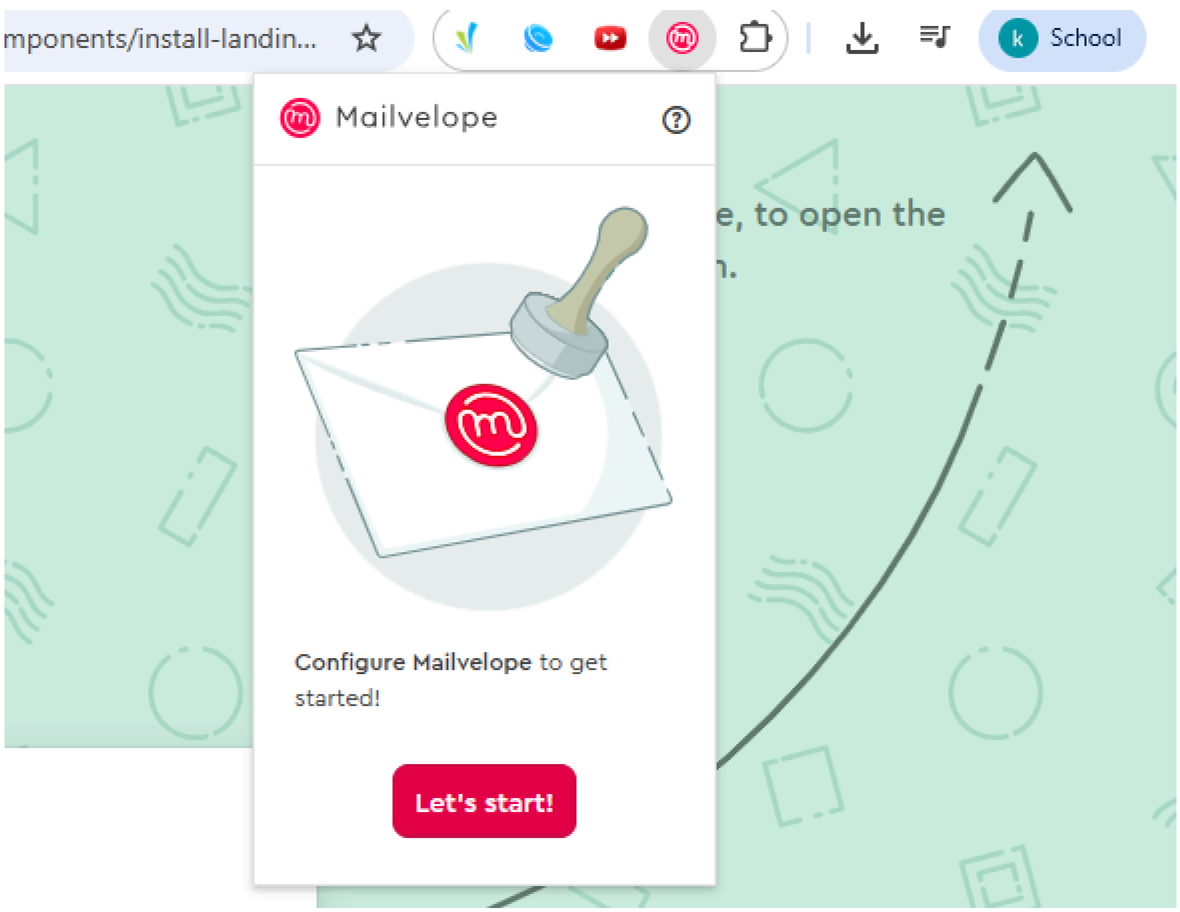
IS A3 | Laiba Fatima 22k-5195 | BSE-7B

Part1_EmailSecurity_22K-5195.pdf

Task 1 Installing extension







I entered the url for the mailvelope extension, then installed it as it was said in the pdf

Task 2 Generating keys

Generate Key

Name

Laiba Fatima

Full name of the key owner

Email

k225195@nu.edu.pk

Advanced >>

Enter Password

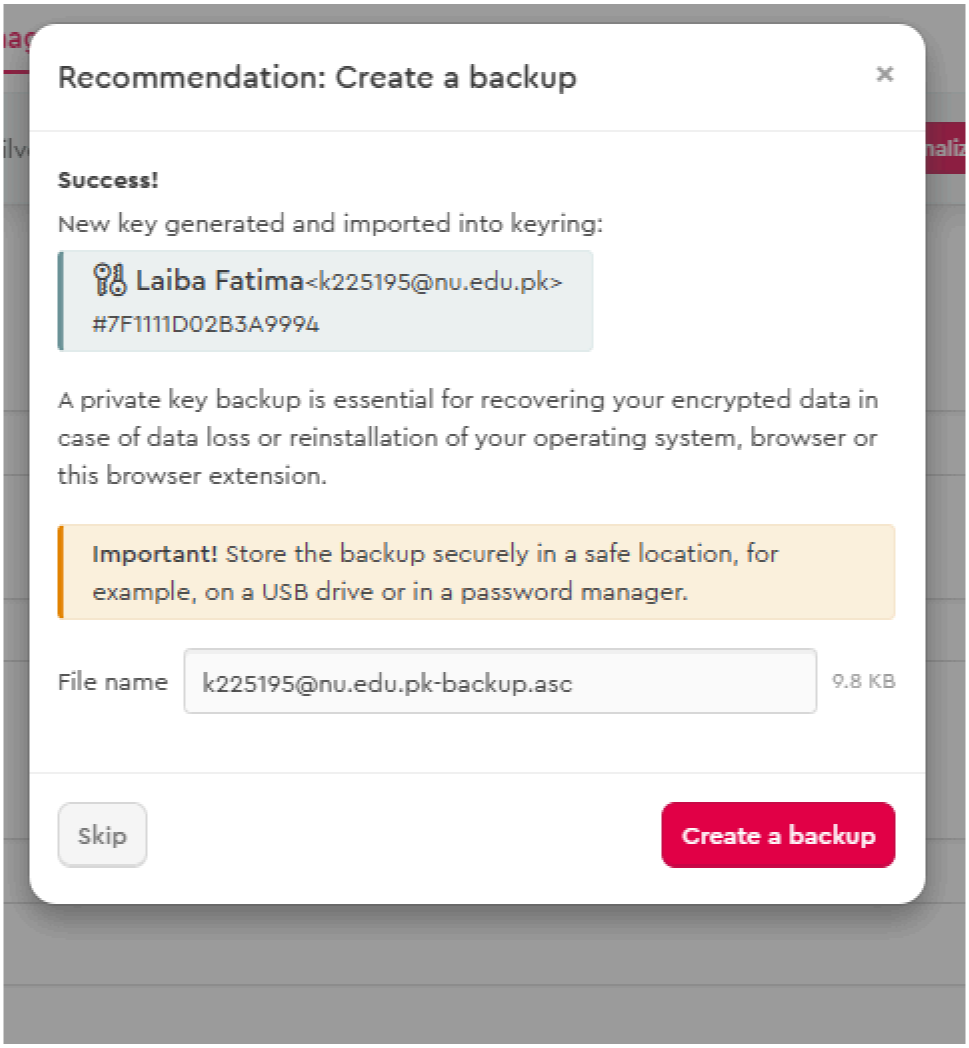
••••••••••••••

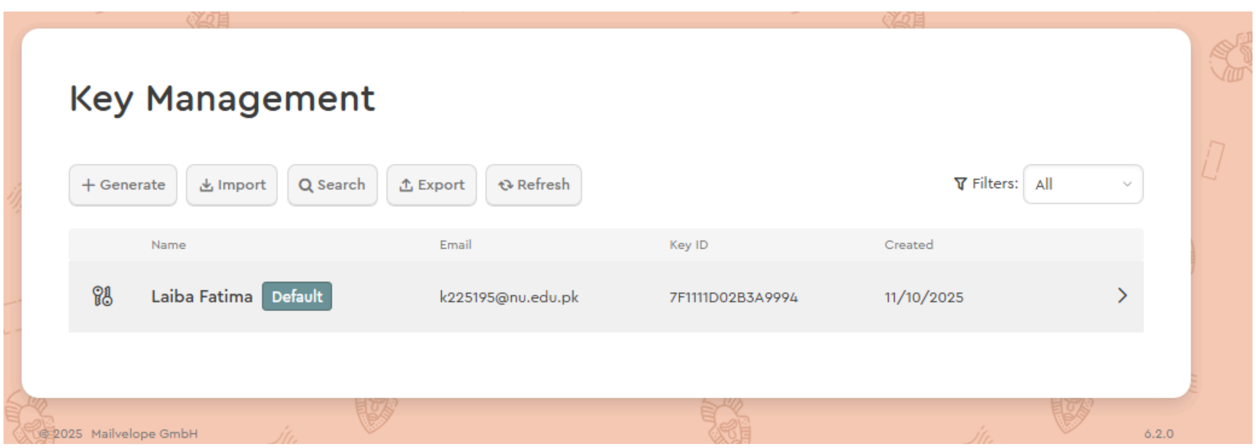
Re-enter Password

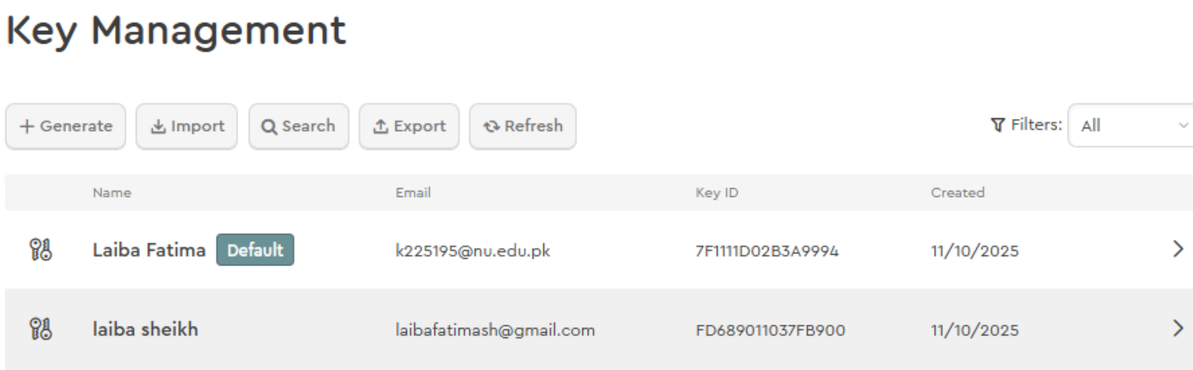
••••••••••••••

☑ Upload public key to Mailvelope Key Server (can be deleted at any time). Learn more

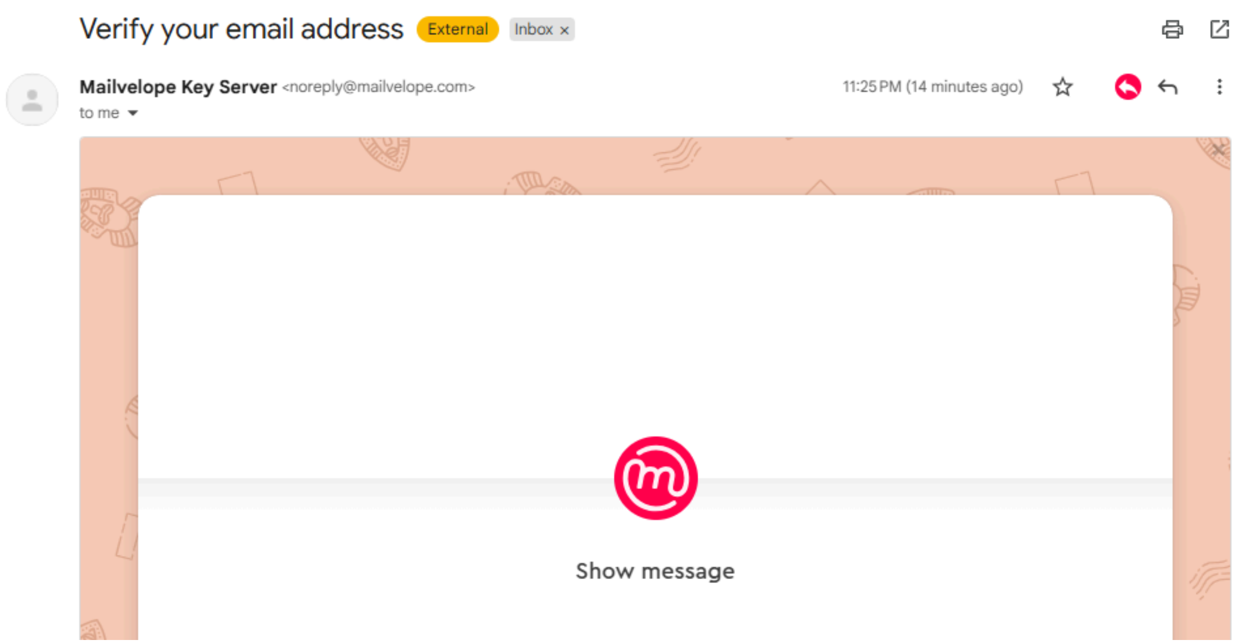
Generate


Recommendation: Create a backup
Success!
New key generated and imported into keyring:
Laiba Fatima<k225195@nu.edu.pk>
#7F1111D02B3A9994
A private key backup is essential for recovering your encrypted data in case of data loss or reinstallation of your operating system, browser or this browser extension.
Important! Store the backup securely in a safe location, for example, on a USB drive or in a password manager.
File name
k225195@nu.edu.pk-backup.asc
9.8 KB
Skip
Create a backup


Key Management
+ Generate
Import
Search
Export
Refresh
Filters:
All
Name
Email
Key ID
Created
Laiba Fatima
Default
k225195@nu.edu.pk
7F1111D02B3A9994
11/10/2025
© 2025 Mailvelope GmbH
6.2.0



I generated key pairs for sender and receiver email accounts
Task 3 Verifying email address



Hello Laiba Fatima,

please verify your email address k225195@nu.edu.pk by clicking on the following link:

https://keys.mailvelope.com/api/v1/key?op=verify&keyId=7f1111d02b3a9994&nonce=70305572a41bfd9a1e6b9abfbc79c4e5

After verification of your email address, your public key is available in our key directory.

You can find more info at keys.mailvelope.com.

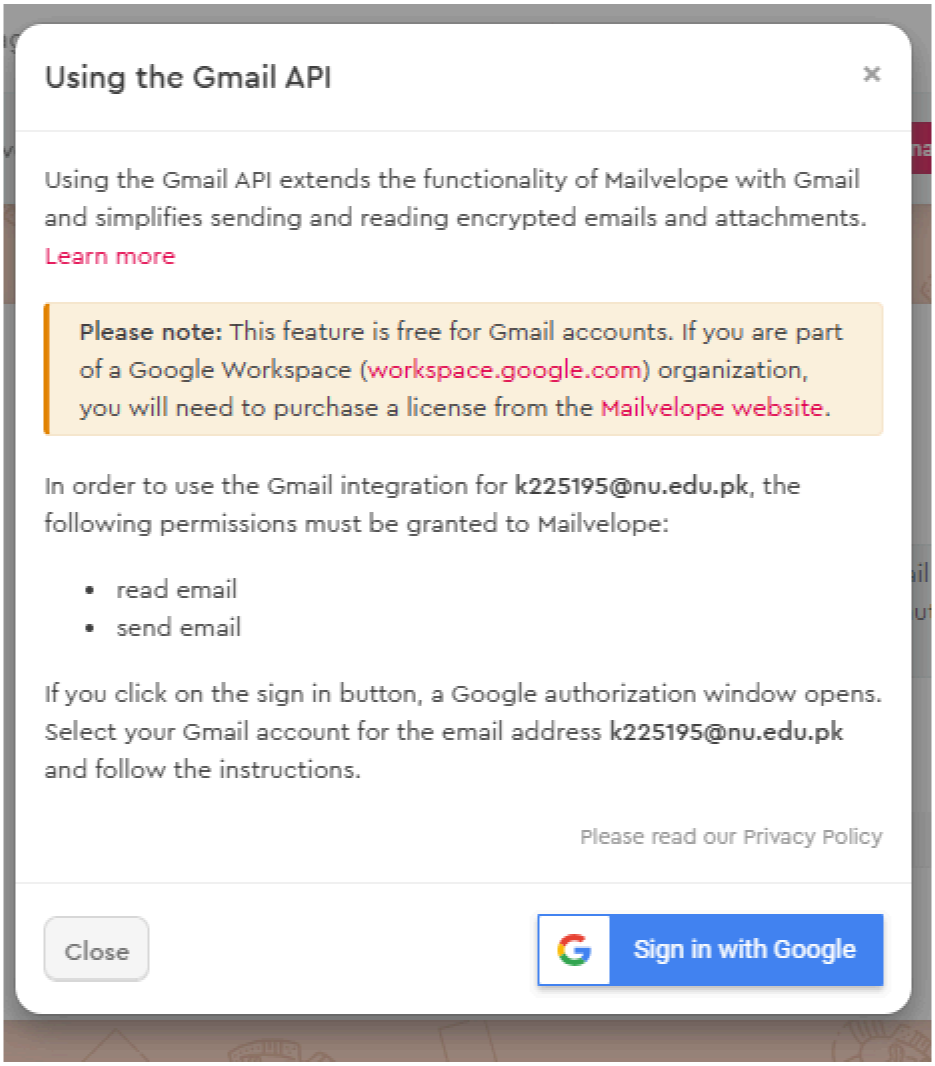
Greetings from the Mailvelope Team

Mailvelope Key Server

Email address k225195@nu.edu.pk successfully verified

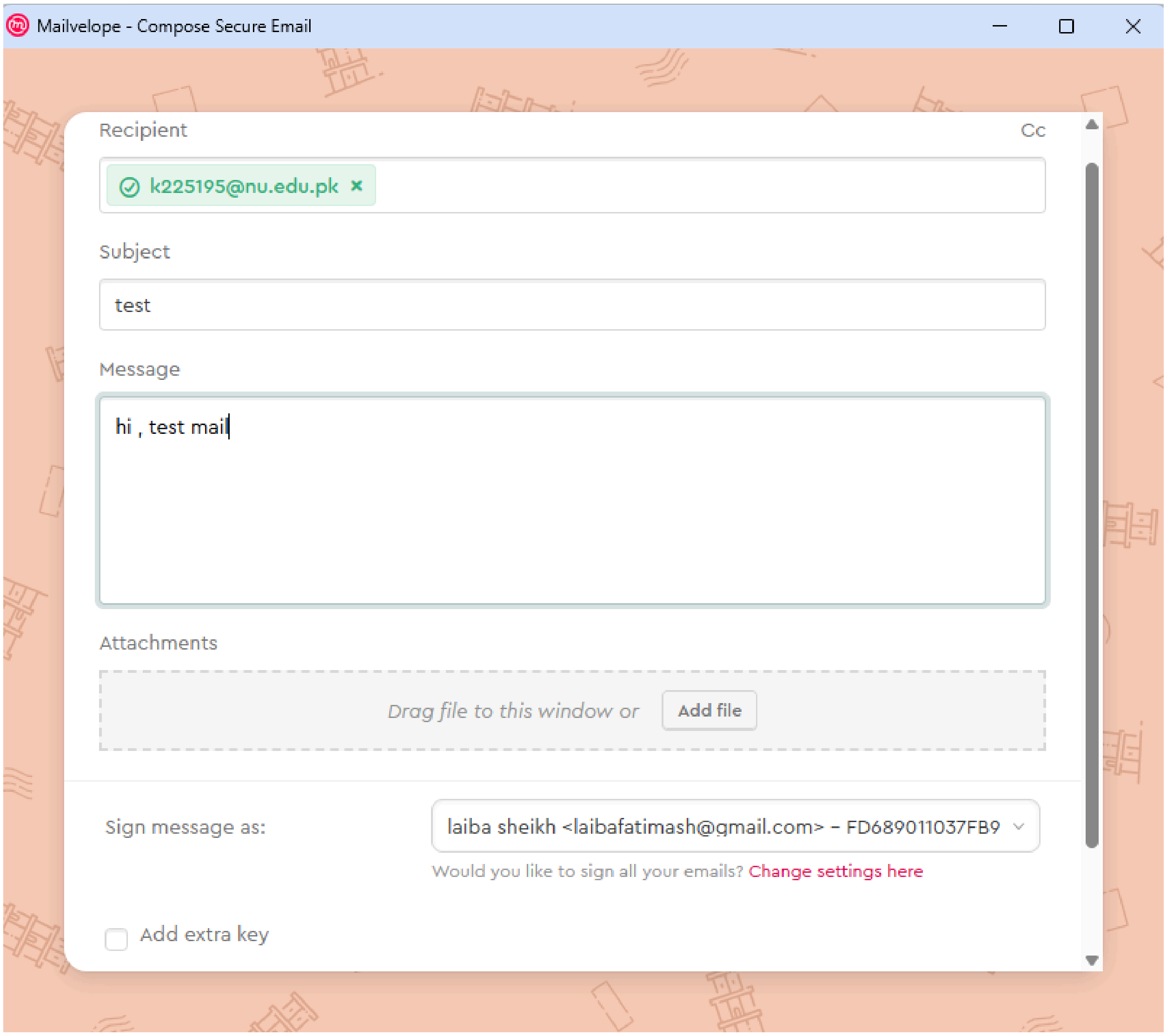
Your public OpenPGP key is now available at the following link: https://keys.mailvelope.com/pks/lookup?op=get&search=k225195@nu.edu.pk

Then i verified the email address for synchronization in sender account
GMAIL API



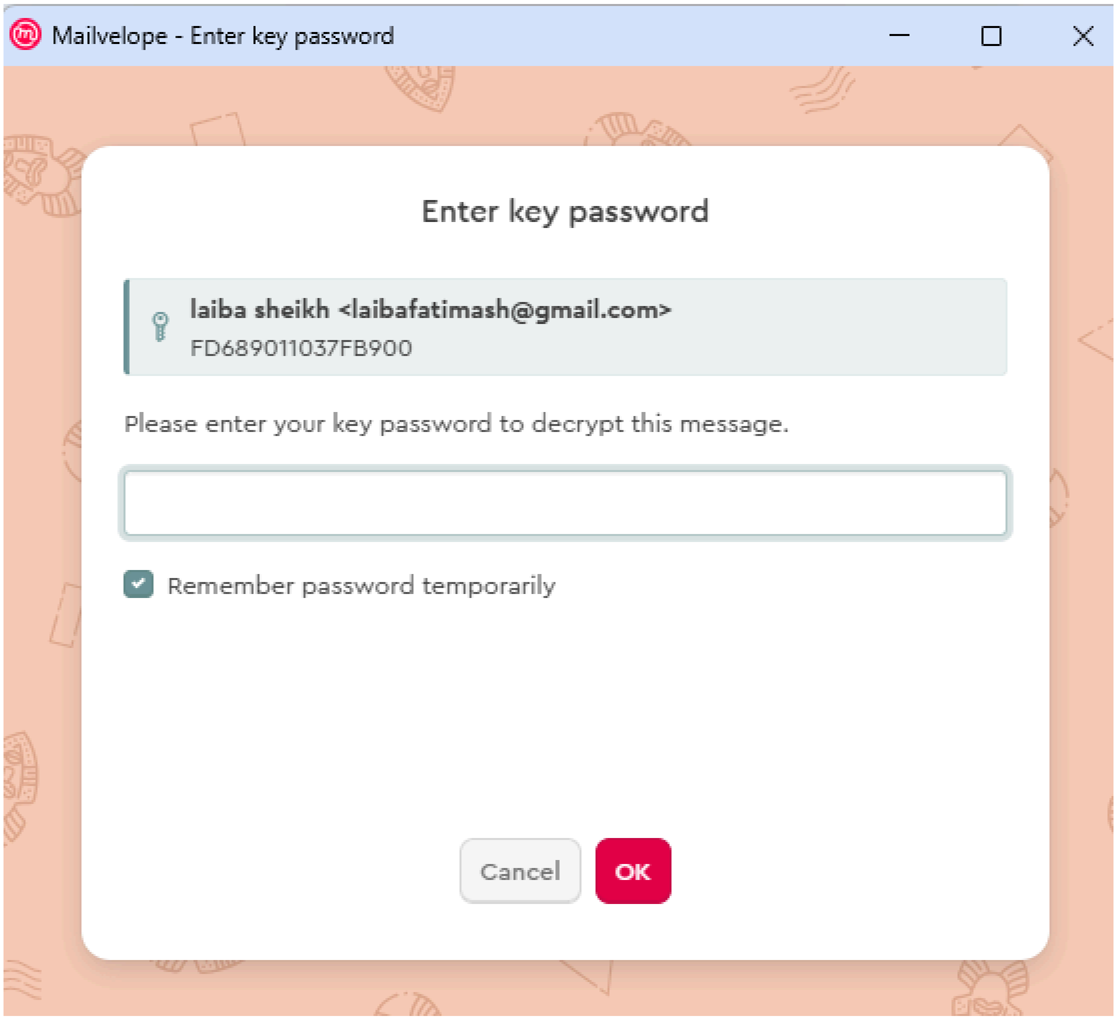
I had trouble setting up gmail api, because it kept telling me to go for a business license.
Then i had to repeat the steps for receiver account and set up the gmail api for receiver account and then it as working fine.

Task 4 Sending email to receiver



I sent the mail from the sender to receiver account

Task 5 Decrypting the mail

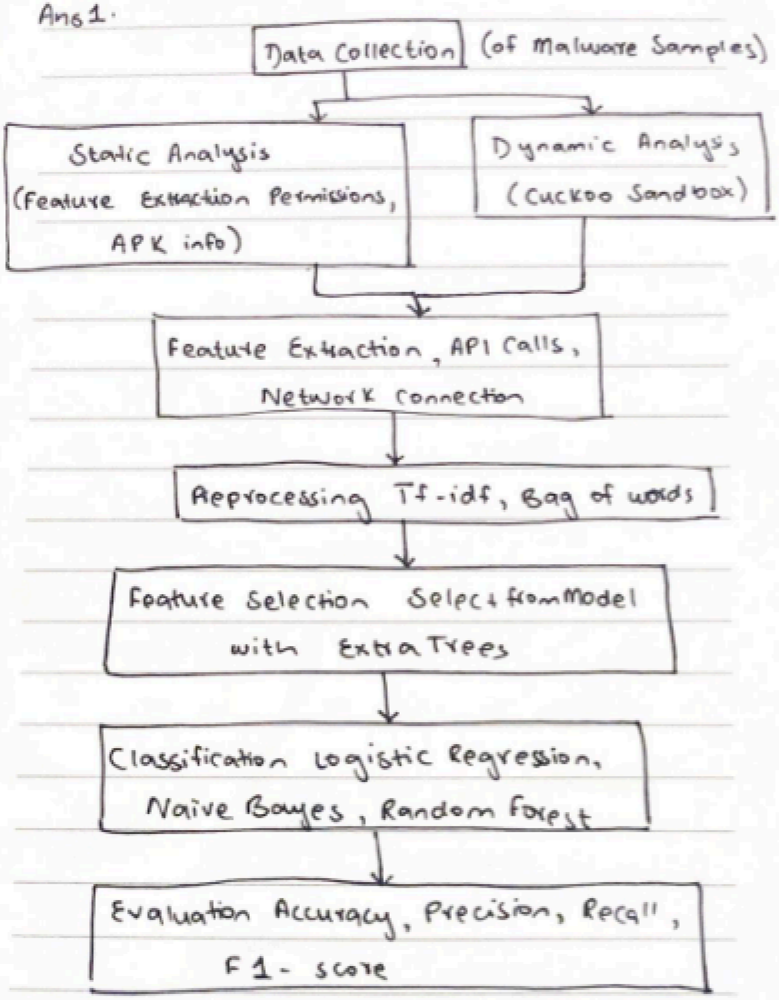


The receiver side decrypted the mail and was able to read

Part2_Research_22K-5195.pdf

Date: Subject:

Ans 1.



Ans 2.

- Nature of Hybrid features : Random Forest model uses a combination of static & dynamic features; which create a high dimensional feature space ; effectively capturing complex feature interactions that Logistic Regression doesn't do.

- Imbalance of Dataset : The malware sample is spread among different families. The model uses bagging & voting from diverse trees so it reduces overfitting on dominant classes & improves minority class recognition compared to Naive Bayes which assumes feature independence or Logistic Regression which can be biased by imbalanced data.

- Adaptability of Ensemble Learners : Random Forest benefits from variance reduction. Each decision tree learns different decision boundaries & combined makes stable & strong predictive power = higher accuracy

Ans3.

i) Static features: email metadata (sender address, subject line, header analysis)

~~ii~~ ii) Dynamic features: behavioral indicators (link clicks, detected payload delivery attempts, execution of embedded scripts)

ii) Challenge 1: Privacy & data sensitivity: emails have sensitive data, collecting & analyzing will raise privacy concerns

Challenge 2: Feature extraction complexity: dynamic behavior harder to capture because interaction might occur outside email (in browser) & be influenced by user action.

Ans4.

i) Strength: Use of VMs adds realism & scalability since multiple samples are analyzed in parallel

Limitation: imbalanced data set will bias classification results towards dominant families causing environment bias

ii) Improvement: Add data augmentation or synthetic minority oversampling techniques to balance dataset across families

Ans 5

i) federated learning so malware training detection happes collaborstively

ii) C : protects data privacy ; keeps data localized
I : improves model robustness by aggregating updates from multiple sources
A : system scales efficiently as more nodes participate without centralized bottleneck improving detection coverage.